

SymPy Bug Card

Bug 57: singularities misses the logarithmic singularity of Chi(x) at zero

Task. Find the complex singularities of the hyperbolic cosine integral.

`singularities(Chi(x), x,  $\mathbb{C}$ )`

SymPy's answer.

`singularities(Chi(x), x,  $\mathbb{C}$ ) =  $\emptyset$ .`

Correct answer. The singularity set must contain 0; for this minimal input, the expected set is $\{0\}$.

Explanation. Near the origin, $\text{Chi}(x) = \gamma + \log(x) + \text{analytic terms}$, so $x = 0$ is a logarithmic singularity. SymPy also evaluates $\text{Chi}(0)$ to `zoo`, and the recorded right-hand limit is $-\infty$, so an empty singularity set is not mathematically equivalent.

Likely root cause. The diagnosis identifies `sympy/calculus/singularities.py` as using a syntactic scan for negative powers, explicit logarithms, and a few inverse hyperbolic atoms; it does not consult Chi's zero value or leading-term hook in the special-function implementation. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-057-singularities-misses-the-logarithmic-singularity-of-chi-x-at/](#)